Python project

Dataset - https://www.kaggle.com/datasets/aliiihussain/amazon-sales-dataset?resource=download

Questions

Part 1: Data Understanding & Cleaning

1. Load the dataset.
2. Display first 10 records.
3. Display last 10 records.
4. Display 5 random records.
5. Check data types of all columns.
6. Find total rows and columns.
7. Check for missing values.
8. Check for duplicate records.

Part 2: New Column Creation

1. Extract Year from Order Date.
2. Extract Month from Order Date.
3. Extract Month Name from Order Date.
4. Extract Day from Order Date.
5. Create a column called **Revenue Category**:
    a. High Revenue (>1000)
    b. Medium Revenue (500–1000)
    c. Low Revenue (<500)
6. Create a column called **Customer Satisfaction**:
    a. Excellent (Rating ≥ 4.5)
    b. Good (Rating 3–4.4)
    c. Poor (<3)

Part 3: Filtering Analysis

1. Find all Electronics orders.
2. Find all orders from Asia.
3. Find orders with Rating greater than 4.
4. Find orders where Revenue exceeds 1000.
5. Find Fashion products sold in Asia.
6. Find Electronics products with Rating above 4.
7. Find orders with Discount greater than 20%.
8. Find orders paid using UPI and having Rating above 4.

Part 4: Business KPIs

1. Calculate Total Revenue.
2. Calculate Total Quantity Sold.
3. Calculate Total Orders.
4. Calculate Average Revenue per Order.

5. Calculate Average Rating.
6. Calculate Average Discount Given.
7. Find Total Number of Products Sold.
8. Find Total Revenue Lost Due to Discounts.

Part 5: Category Analysis

1. Total Revenue by Product Category.
2. Total Quantity Sold by Product Category.
3. Average Rating by Product Category.
4. Average Discount by Product Category.
5. Highest Revenue Generating Category.
6. Lowest Revenue Generating Category.
7. Highest Selling Category.
8. Lowest Selling Category.

Part 6: Region Analysis

1. Total Revenue by Region.
2. Total Quantity Sold by Region.
3. Average Rating by Region.
4. Highest Revenue Region.
5. Lowest Revenue Region.
6. Region with Maximum Orders.
7. Region with Minimum Orders.

Part 7: Payment Method Analysis

1. Total Revenue by Payment Method.
2. Total Orders by Payment Method.
3. Average Rating by Payment Method.
4. Most Preferred Payment Method.
5. Least Preferred Payment Method.

Part 8: Discount Analysis

1. Revenue Generated by Each Discount Level.
2. Average Rating by Discount Level.
3. Which Discount Percentage Generated Maximum Revenue?
4. Which Discount Percentage Received Highest Ratings?

Part 9: Customer Review Analysis

1. Average Rating by Category.
2. Average Rating by Region.
3. Highest Rated Category.
4. Lowest Rated Category.
5. Categories with More Than 100 Average Reviews.

Part 10: Date Analysis

1. Total Revenue by Year.
2. Total Orders by Year.

3. Total Revenue by Month.
4. Total Orders by Month.
5. Month with Highest Revenue.
6. Month with Lowest Revenue.
7. Month with Highest Sales.
8. Month with Lowest Sales.
9. Day-wise Revenue Analysis.
10. Day-wise Sales Analysis.

Part 11: Top & Bottom Analysis

1. Top 10 Products by Revenue.
2. Bottom 10 Products by Revenue.
3. Top 10 Products by Quantity Sold.
4. Bottom 10 Products by Quantity Sold.
5. Top 5 Categories by Revenue.
6. Bottom 5 Categories by Revenue.
7. Top 5 Regions by Revenue.
8. Bottom 5 Regions by Revenue.

Part 12: Visualizations

Bar Charts

1. Revenue by Product Category.
2. Revenue by Region.
3. Revenue by Payment Method.

Horizontal Bar Charts

1. Top 10 Products by Revenue.
2. Top 10 Products by Quantity Sold.

Pie Charts

1. Revenue Contribution by Category.
2. Orders by Payment Method.

Histograms

1. Distribution of Ratings.
2. Distribution of Revenue.

Scatter Plots

1. Rating vs Revenue.
2. Discount Percentage vs Revenue.

Line Charts

1. Monthly Revenue Trend.
2. Monthly Sales Trend.
3. Daily Revenue Trend
4. Daily Sales Trend

Sample new column creation code

```
#Create Revenue Category Column
#Requirement
#High Revenue (>1000) Medium Revenue (500–1000)Low Revenue (<500)
def revenue_category(revenue):
    if revenue > 1000:
        return "High Revenue"
    elif revenue >= 500:
        return "Medium Revenue"
    else:
        return "Low Revenue"

df['Revenue_Category'] = df['total_revenue'].apply(revenue_category)
print(df[['total_revenue','Revenue_Category']].head())
```